



Signals & Systems

Detailed Solutions

GATE INSTRUMENTATION ENGINEERING

PREPFUSION

Previous Year Questions Solutions



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SOLUTIONS



Basics of Signals

Topics

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Q1.26.1

NAT

2026

2M

Ans: 2.5



Given:

$$f(t) = \begin{cases} 1, & t \in [0, 2] \\ -t + 3, & t \in [2, 3] \\ 0, & \text{otherwise} \end{cases}$$

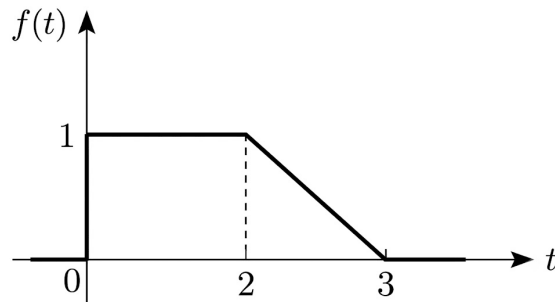


Fig. Solution Q1.26.1

$$\int_{-\infty}^{\infty} f(t) dt = 2.5$$

$$\text{Area} = 2 + \frac{1}{2} \times 1 \times 1 = 2.5$$

Hence, the correct answer is **2.5**.

2.5

Q1.26.1

MCQ

2026

2M

Ans: B



We need to determine

$$y(t) = 1 - x(4 - t).$$

The transformations are applied in the following order:

1. Time reversal:

$$x(t) \rightarrow x(-t)$$

2. Shift right by 4:

$$x(-t) \rightarrow x(4 - t)$$

3. Amplitude inversion and upward shift by 1:

$$y(t) = 1 - x(4 - t)$$

Break points of $x(t)$:

$$t = -3, -2, -1, 3, 5.$$

After the transformation $t \rightarrow 4 - t$,

$$4 - (-3) = 7, \quad 4 - (-2) = 6, \quad 4 - (-1) = 5, \quad 4 - 3 = 1, \quad 4 - 5 = -1.$$

Hence, the transformed signal has break points at

$$t = -1, 1, 5, 6, 7.$$

For $-1 \leq t \leq 1$, $x(4-t)$ corresponds to the ramp portion of $x(t)$, varying from -2 to 0 . Therefore,

$$y(t) = 1 - x(4-t)$$

varies linearly from

$$1 - (-2) = 3$$

to

$$1 - 0 = 1.$$

Thus, a decreasing ramp is obtained between $t = -1$ and $t = 1$.

For $1 < t < 5$,

$$x(4-t) = 1 \Rightarrow y(t) = 1 - 1 = 0.$$

For $5 < t < 6$,

$$x(4-t) = 0 \Rightarrow y(t) = 1.$$

For $6 < t < 7$, $x(4-t)$ is the negative ramp from 0 to -1 , giving a descending ramp from 1 to 0 .

Thus, the resulting waveform matches option **(B)**.

(B)

Q1.26.3

MCQ

2026

1M

Ans: D

● ○ ○

Given:

$$x(t) = 4 \sin(15\pi t) + 7 \cos(4\pi t)$$

For $x_1(t)$, $\omega_1 = 15\pi$,

$$T_1 = \frac{2\pi}{15\pi} = \frac{2}{15}$$

For $x_2(t)$, $\omega_2 = 4\pi$,

$$T_2 = \frac{2\pi}{4\pi} = \frac{1}{2}$$

The sum of periodic signals is periodic if the ratio of their periods is rational.

$$\frac{T_1}{T_2} = \frac{2/15}{1/2} = \frac{4}{15}$$

This is rational, so the signal is periodic.

$$T_0 = \text{LCM}[T_1, T_2]$$

$$T_0 = \text{LCM}\left[\frac{2}{15}, \frac{1}{2}\right]$$

$$T_0 = \frac{\text{LCM}(2, 1)}{\text{HCF}(15, 2)} = \frac{2}{1}$$

$$T_0 = 2 \text{ seconds}$$

Hence, the correct option is **(D)**.

(D) 2

Q1.25.1

MCQ

2025

1M

Ans: B

● ○ ○

Given: $x(t) = e^{j9t} + e^{j5t}$ where $j = \sqrt{-1}$,

$$|x(t)| = |e^{j9t} + e^{j5t}|$$

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$x(t) = \cos 9t + j \sin 9t + \cos 5t + j \sin 5t$$

$$|x(t)| = \sqrt{(\cos 5t + \cos 9t)^2 + (\sin 5t + \sin 9t)^2}$$

$$|x(t)| = \sqrt{\cos^2 5t + \cos^2 9t + 2 \cos 5t \cos 9t + \sin^2 5t + \sin^2 9t + 2 \sin 5t \sin 9t}$$

$$|x(t)| = \sqrt{(1 + 1 + 2 \cos 5t \cos 9t + 2 \sin 5t \sin 9t)}$$

$$|x(t)| = \sqrt{2 + 2 \cos 4t}$$

$$|x(t)| = \sqrt{2(1 + \cos 4t)}$$

$$|x(t)| = \sqrt{2(1 + 2 \cos^2 2t - 1)}$$

$$|x(t)| = \sqrt{2 \times 2 \cos^2 2t}$$

$$|x(t)| = |2 \cos 2t|$$

The fundamental period of $\cos(2t)$ is,

$$T = \frac{2\pi}{2} = \pi$$

But for $|\cos(2t)|$, the period is halved.

$$T = \frac{\pi}{2} \text{ sec}$$

(B) $\frac{\pi}{2}$

Q1.23.1

NAT

2023

1M

Ans: -3

● ○ ○

Given:

$$x(n) = u(-n + 5) - u(n + 3)$$

The signal $x(n)$ can be plotted as shown below,

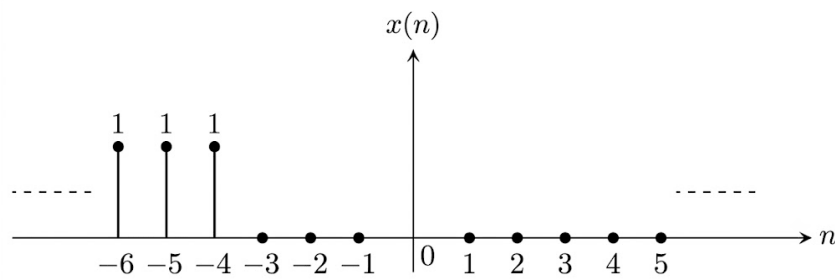


Fig. Solution Q1.23.1

So, the smallest value of n is -3 for $x(n)$ to be zero.

Hence, the correct answer is -3 .

-3.

Q1.23.2

MCQ

2023

1M

Ans: C

● ○ ○

Theory: For an LTI system, the steady-state response to a sinusoidal input is determined by the magnitude and phase of the frequency response $H(j\omega)$.

The transfer function is

$$H(s) = \frac{s - \pi}{s + \pi}.$$

Since

$$y(t) = \sin(\pi t),$$

the excitation frequency is

$$\omega = \pi \text{ rad/s.}$$

Evaluating the frequency response,

$$H(j\omega) = \frac{j\omega - \pi}{j\omega + \pi}.$$

At $\omega = \pi$,

$$H(j\pi) = \frac{j\pi - \pi}{j\pi + \pi}.$$

The magnitude is

$$|H(j\pi)| = \frac{\sqrt{\pi^2 + \pi^2}}{\sqrt{\pi^2 + \pi^2}} = 1.$$

The phase is

$$\angle H(j\omega) = 180^\circ - 2 \tan^{-1}\left(\frac{\omega}{\pi}\right).$$

Substituting $\omega = \pi$,

$$\angle H(j\pi) = 180^\circ - 2 \tan^{-1}(1) = 180^\circ - 90^\circ = 90^\circ.$$

Since

$$Y(j\omega) = H(j\omega)X(j\omega),$$

the output leads the input by 90° . Therefore, the input must lag the output by 90° .

Therefore,

$$(C)x(t) = \sin\left(\pi t - \frac{\pi}{2}\right)u(t)$$

Key Points

- Evaluate the transfer function at the excitation frequency to obtain the steady-state gain and phase.
- The output phase equals the input phase plus the phase of $H(j\omega)$.

- Since $|H(j\pi)| = 1$, only the phase shift affects the input signal.

Mistakes to be avoided

- Do not substitute $s = \pi$; use $s = j\omega$.
- Do not confuse whether the input leads or lags the output; use $Y = HX$.
- Verify both magnitude and phase before selecting the input waveform.

Q1.23.3

NAT

2023

1M

Ans: 25



Theory: Time scaling changes the period of a continuous-time periodic signal according to $T' = \frac{T}{|a|}$ for $x(at)$.
Given,

$$y(t) = x(4t),$$

and the fundamental period of $x(t)$ is

$$T_x = 100 \text{ s.}$$

If $x(t)$ has fundamental period T , then

$$x(at)$$

has fundamental period

$$T' = \frac{T}{|a|}.$$

Hence,

$$T_y = \frac{100}{4} = 25 \text{ s.}$$

Therefore,

$$\boxed{25}$$

Key Points

- Time compression by a factor of a reduces the period by the same factor.
- For $x(at)$, the new period is $T/|a|$.
- Always use the magnitude $|a|$ while computing the new period.

Mistakes to be avoided

- Do not multiply the period by the scaling factor; divide by $|a|$.
- Do not ignore the absolute value of the time-scaling factor.
- Ensure that the given period is the fundamental period before applying the formula.

Q1.23.4

MCQ

2023

2M

Ans: C



Theory: For a sinusoidal input, the steady-state output of an LTI system is obtained by evaluating the frequency response $H(j\omega)$ at the input frequency.

Given,

$$h(t) = \delta(t) + 0.5\delta(t - 4),$$

and

$$x(t) = \cos\left(\frac{7\pi}{4}t\right).$$

The frequency response is

$$H(j\omega) = 1 + 0.5e^{-j4\omega}.$$

The input frequency is

$$\omega = \frac{7\pi}{4}.$$

Hence,

$$H\left(j\frac{7\pi}{4}\right) = 1 + 0.5e^{-j7\pi} = 1 + 0.5(-1) = 0.5.$$

Since the frequency response is purely real,

$$\angle H\left(j\frac{7\pi}{4}\right) = 0^\circ,$$

there is no phase shift.

Therefore,

$$y(t) = H\left(j\frac{7\pi}{4}\right)x(t) = 0.5 \cos\left(\frac{7\pi}{4}t\right).$$

Hence,

$$(A)y(t) = 0.5 \cos\left(\frac{7\pi}{4}t\right)$$

Key Points

- The frequency response is the Fourier transform of the impulse response.
- A delayed impulse contributes a factor of $e^{-j\omega\tau}$ in the frequency response.
- For sinusoidal inputs, evaluate $H(j\omega)$ at the input frequency.

Mistakes to be avoided

- Do not substitute $e^{-j7\pi} = 1$; since 7 is odd, $e^{-j7\pi} = -1$.
- Do not perform time-domain convolution when the frequency-response method is simpler.
- Check whether $H(j\omega)$ introduces any phase shift before writing the output.

Q1.21.1

MCQ

2021

1M

Ans: C



Given:

$$x(t) = \sin \sqrt{2\pi}t$$

For a non-DC signal to be periodic, there must exist a constant T , such that,

$$x(t \pm T) = x(t) \quad \dots (i)$$

Equation (i) can be satisfied only if the power of t in $x(t)$ is 1.

As here the power of t in $x(t)$ is

$$\frac{1}{2}.$$

So, the signal is not periodic.

Hence, the correct option is (C).

(C) Not periodic

Q1.20.1

MCQ

2020

1M

Ans: B



Given:

$$x[n] = \sin(2\pi n)u[n],$$

where $u[n] = 1$ for $n \geq 0$.

$$x[0] = \sin 2\pi(0) u[0] = 0$$

$$x[1] = \sin 2\pi(1) u[1] = 0$$

$$x[2] = \sin 2\pi(2) u[2] = 0$$

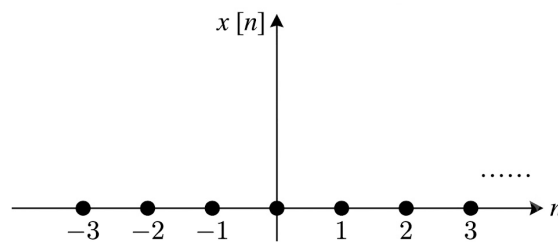


Fig. Solution Q1.20.1

$x[n]$ is a constant (DC) signal of zero value where it repeats its value zero with each value of n . So, the time period = 1.

Hence, the correct option is (B).

(B) 1

Q1.20.2

NAT

2020

1M

Ans: 0



Theory: Convolution with the unit impulse sequence leaves the original sequence unchanged.

Given,

$$g[n] = \delta[n].$$

Hence,

$$y[n] = h[n] \otimes g[n] = h[n] \otimes \delta[n] = h[n].$$

The sequence $h[n]$ is

$$h[n] = \begin{cases} 1, & n = 0, 3, 6, 9, \dots \\ 0, & \text{otherwise.} \end{cases}$$

Since 2 is not a multiple of 3,

$$h[2] = 0.$$

Therefore,

$$y[2] = h[2] = 0.$$

y[2]=0

Key Points

- The sequence $g[n]$ is the unit impulse sequence $\delta[n]$.
- Convolution with $\delta[n]$ satisfies $x[n] \otimes \delta[n] = x[n]$.

- The sequence $h[n]$ is nonzero only at indices $0, 3, 6, \dots$

Mistakes to be avoided

- Do not perform lengthy convolution when one sequence is $\delta[n]$.
- Do not confuse convolution with pointwise multiplication.
- Check whether the required index belongs to the set of nonzero indices before evaluating the sequence.

Q1.19.1**NAT****2019****2M****Ans: 6**

Given:

$$x[n] = e^{j(\frac{5\pi}{3})n} + e^{j(\frac{\pi}{4})n}$$

For down-sampling by 4,

$$n = 4n$$

$$x[4n] = e^{j(\frac{5\pi}{3})4n} + e^{j(\frac{\pi}{4})4n}$$

$$\omega_1 = \frac{20\pi}{3}, \quad \omega_2 = \pi$$

$$N_1 = \frac{2\pi k}{20\pi/3} = \frac{3k}{10}$$

Putting $k = 10$,

$$N_1 = 3$$

$$N_2 = \frac{2\pi k}{\pi} = 2k$$

Putting $k = 1$,

$$N_2 = 2$$

$$(N_1, N_2) = (3, 2)$$

$$\text{Fundamental period} = \text{LCM}(N_1, N_2)$$

$$\text{Fundamental period} = 6$$

Hence, the fundamental period of the down-sampled signal $x_d[n]$ is 6.

6

Q1.18.1

MCQ

2018

1M

Ans: A



Given:

$$x(t) = \begin{cases} 1, & |t| \leq 2 \\ 0, & |t| > 2 \end{cases}$$

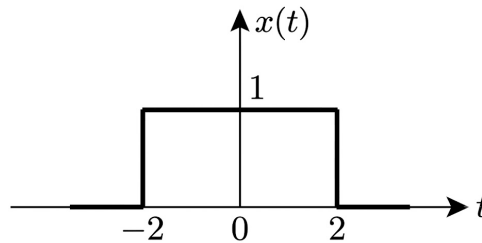


Fig. Solution Q1.18.1

$$I = \int_0^5 2x(t-3)\delta(t-4) dt$$

$$I = 2x(4-3) \int_0^5 \delta(t-4) dt$$

Since, area under the impulse function is 1,

$$I = 2x(1) \times 1 = 2$$

Hence, the correct option is (A).

(A) 2

Q1.18.2

MCQ

2018

1M

Ans: D



Given:

$$z(t) = x(-t) + y(2t + 1)$$

where, $x(t)$ and $y(t)$ are periodic signals with period 3 secs.

Period of $x(-t)$ is 3 sec.

$$\text{Period of } y\left[2\left(t + \frac{1}{2}\right)\right] = \frac{3}{2} = 1.5$$

Now, period of $z(t)$ is given by,

$$T = \text{LCM}(3, 1.5) = 3$$

Hence, the correct option is (D).

(D) 3

Key Points

- Time inversion and time shifting do not change the time period of a signal.
- If signal $x(t)$ has period T_0 , after time scaling the signal $x(t)$ by factor k , i.e., $x(kt)$,

$$\text{Time period} = \frac{T_0}{k}.$$

Q1.17.1 NAT 2017 1M Ans: 1 ☒ ☐ ☐

Given:

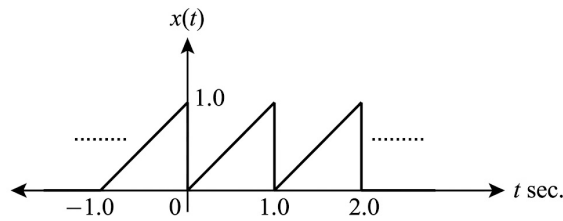


Fig. Solution Q1.17.1

Fundamental period,

$$T_0 = 1 \text{ sec}$$

Fundamental frequency,

$$f_0 = \frac{1}{T_0} = 1 \text{ Hz}$$

Hence, fundamental frequency of signal $x(t)$ is 1 Hz.

1

Q1.16.1 NAT 2016 2M Ans: 8 ☒ ☐ ☐

Given:

$$x(n) = \sin\left(\frac{301}{4}\pi n\right)$$

where,

$$\omega_0 = \frac{301}{4}\pi \text{ rad/sec}$$

Fundamental time period of discrete-time signal is given by,

$$N_0 = m \times \frac{2\pi}{\omega_0}$$

where, m is the minimum integer number to de-rationalize time period N_0 .

$$N_0 = m \times \frac{2\pi}{301\pi/4}$$

$$N_0 = m \times \frac{8}{301}$$

For $m = 301$,

$$N_0 = 8$$

Hence, the fundamental period N_0 is 8.

8

Q1.15.1**NAT****2015****2M****Ans: 6**

● ○ ○

Given:

$$x(t) = 2 \cos\left(\frac{2\pi}{3}t\right) + \cos(\pi t)$$

Method Method 1

$$\omega_1 = \frac{2\pi}{3} \text{ rad/sec}, \quad T_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{2\pi/3} = 3 \text{ sec}$$

$$\omega_2 = \pi \text{ rad/sec}, \quad T_2 = \frac{2\pi}{\omega_2} = \frac{2\pi}{\pi} = 2 \text{ sec}$$

Fundamental time period of T_1 and T_2 is calculated as,

	$\times 1$	$\times 2$	$\times 3$
$T_1 = 3$	3	6	9
$T_2 = 2$	2	4	6

Fundamental time period,

$$T_0 = 6 \text{ sec}$$

Hence, fundamental time period of the signal is 6 sec.

Method Method 2

$$\omega_1 = \frac{2\pi}{3} \text{ rad/sec}$$

$$\omega_2 = \pi \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2)$$

$$\omega_0 = \text{HCF}\left(\frac{2\pi}{3}, \pi\right) = \frac{\pi}{3}$$

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi/3} = 6$$

Hence, fundamental time period of the signal is 6 sec.

6

Q1.13.1**MCQ****2013****1M****Ans: A**

● ○ ○

Given:

$$v(t) = 30 \sin 100t + 10 \cos 300t + 6 \sin\left(500t + \frac{\pi}{4}\right)$$

$$\omega_1 = 100 \text{ rad/sec}$$

$$\omega_2 = 300 \text{ rad/sec}$$

$$\omega_3 = 500 \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3)$$

$$\omega_0 = \text{HCF}(100, 300, 500)$$

$$\omega_0 = 100 \text{ rad/sec}$$

Hence, the correct option is (A).

(A) 100

Q1.11.1

MCQ

2011

1M

Ans: C

☒ ☐ ☐

Given:

$$x(n) = \sin \omega_0 n$$

For a discrete-time signal to be periodic, the necessary condition is given by,

$$\frac{m}{N_0} = \frac{\omega_0}{2\pi} = \frac{p}{q} = \text{Ratio of integers.}$$

From the given options, only option (C) satisfies the above condition.

Hence, the correct option is (C).

(C) $\frac{2\pi}{\omega_0}$ to be a ratio of integers

Q1.11.2

MCQ

2011

1M

Ans: A

☒ ☐ ☐

Given:

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} \delta(1-2t) dt$$

Method Method 1

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} \times \frac{1}{2} \delta\left(t - \frac{1}{2}\right) dt$$

$$I = \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{2}\right)^2 e^{-(1/2)^2/2} \delta\left(t - \frac{1}{2}\right) dt$$

$$I = \frac{1}{8\sqrt{2\pi}} e^{-1/8} \int_{-\infty}^{\infty} \delta\left(t - \frac{1}{2}\right) dt$$

Since, area under the impulse function is 1,

$$I = \frac{1}{8\sqrt{2\pi}} e^{-1/8}$$

Hence the correct option is (A).

Method Method 2

Using integration property,

$$\int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0)$$

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} \delta(1 - 2t) dt$$

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} \times \frac{1}{2} \delta\left(t - \frac{1}{2}\right) dt$$

$$I = \frac{1}{2\sqrt{2\pi}} \left(\frac{1}{2}\right)^2 e^{-(1/2)^2/2}$$

$$I = \frac{1}{8\sqrt{2\pi}} e^{-1/8}$$

Hence the correct option is (A).

(A) $\frac{1}{8\sqrt{2\pi}} e^{-1/8}$

Q1.10.1

MCQ

2010

1M

Ans: B



Given:

$$I = \int_{-\infty}^{\infty} \delta\left(t - \frac{\pi}{6}\right) 6 \sin t dt$$

Method Method 1

$$I = \int_{-\infty}^{\infty} 6 \sin\left(\frac{\pi}{6}\right) \delta\left(t - \frac{\pi}{6}\right) dt$$

$$I = 6 \times \frac{1}{2} \int_{-\infty}^{\infty} \delta\left(t - \frac{\pi}{6}\right) dt$$

Since, area under the impulse function is 1,

$$I = 3 \times 1 = 3$$

Hence the correct option is (B).

Method Method 2

Using integration property,

$$\int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0)$$

$$I = \int_{-\infty}^{\infty} \delta\left(t - \frac{\pi}{6}\right) 6 \sin t dt$$

$$I = 6 \sin t \Big|_{t=\pi/6} = 3$$

Hence, the correct option is (B).

(B) 3

Q1.09.1**MCQ****2009****1M****Ans: C**

Given:

$$x(t) = 2 \sin 2\pi t + 3 \sin 3\pi t$$

where,

$$\omega_1 = 2\pi \text{ rad/sec}$$

and

$$\omega_2 = 3\pi \text{ rad/sec}$$

Fundamental angular frequency,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2) = \text{HCF}(2\pi, 3\pi)$$

$$\omega_0 = \pi \text{ rad/sec}$$

Fundamental time period,

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi} = 2 \text{ sec}$$

Hence, the correct option is (C).

(C) 2 sec

Q1.08.1**MCQ****2008****1M****Ans: D**

Given:

$$x(n) = e^{j\left(\frac{5\pi}{6}\right)n}$$

where,

$$\omega_0 = \frac{5\pi}{6} \text{ rad/sec}$$

For a discrete-time signal to be periodic,

$$\frac{m}{N_0} = \frac{\omega_0}{2\pi} = \text{Ratio of integers}$$

$$N_0 = \frac{2\pi}{\omega_0} \times m = \frac{2\pi}{5\pi/6} \times m$$

$$N_0 = \frac{12}{5} \times m$$

where, m is the minimum integer number to de-rationalize the time period N_0 .For $m = 5$,

$$N_0 = 12$$

Hence, the correct option is (D).

(D) 12

Q1.07.1

MCQ

2007

2M

Ans: C



Given:

$$x(n) = \left(\frac{1}{3}\right)^n u(n)$$

where,

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

and

$$y(n) = x(-n), \quad -\infty < n < \infty$$

$$\begin{aligned} \sum_{n=-\infty}^{\infty} y(n) &= \sum_{n=-\infty}^{\infty} x(-n) \\ \sum_{n=-\infty}^{\infty} y(n) &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{3}\right)^{-n} u(-n) \\ \sum_{n=-\infty}^{\infty} y(n) &= \sum_{n=-\infty}^0 \left(\frac{1}{3}\right)^{-n} \\ \sum_{n=-\infty}^{\infty} y(n) &= 1 + \left(\frac{1}{3}\right)^1 + \left(\frac{1}{3}\right)^2 + \dots \\ \sum_{n=-\infty}^{\infty} y(n) &= \frac{1}{1 - \frac{1}{3}} = \frac{3}{2} \end{aligned}$$

Hence, the correct option is (C).

$$(C) \frac{3}{2}$$

Q1.07.2

MCQ

2007

1M

Ans: A



Given:

$$x(t) = (1 + 0.5 \cos 40\pi t) \cos 200\pi t$$

$$x(t) = \cos 200\pi t + \frac{0.5}{2} [\cos 240\pi t + \cos 160\pi t]$$

$$x(t) = \cos 200\pi t + 0.25 \cos 240\pi t + 0.25 \cos 160\pi t$$

$$x(t) = 0.25 \cos 160\pi t + \cos 200\pi t + 0.25 \cos 240\pi t$$

where,

$$\omega_1 = 160\pi \text{ rad/sec}$$

$$\omega_2 = 200\pi \text{ rad/sec}$$

$$\omega_3 = 240\pi \text{ rad/sec}$$

Fundamental angular frequency,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3)$$

$$\omega_0 = \text{HCF}(160\pi, 200\pi, 240\pi)$$

$$\omega_0 = 40\pi \text{ rad/sec}$$

Fundamental frequency,

$$f_0 = \frac{\omega_0}{2\pi} = \frac{40\pi}{2\pi} = 20 \text{ Hz}$$

Hence, the correct option is (A).

(A) 20

Q1.06.1

MCQ

2006

2M

Ans: A



Given:

$$x(t) = \cos(1.2\pi t) + \cos(2\pi t) + \cos(2.8\pi t)$$

$$\omega_1 = 1.2\pi \text{ rad/sec} = \frac{6\pi}{5} \text{ rad/sec}$$

$$\omega_2 = 2\pi \text{ rad/sec}$$

$$\omega_3 = 2.8\pi \text{ rad/sec} = \frac{14\pi}{5} \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3)$$

$$\omega_0 = \text{HCF}\left(\frac{6\pi}{5}, \frac{2\pi}{1}, \frac{14\pi}{5}\right)$$

$$\omega_0 = \frac{\text{HCF}[6\pi, 2\pi, 14\pi]}{\text{LCM}[5, 1, 5]}$$

$$\omega_0 = \frac{2\pi}{5} \text{ rad/sec}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{\frac{2\pi}{5}}{2\pi} = \frac{1}{5} = 0.2 \text{ Hz}$$

Hence, the correct option is (A).

(A) 0.2 Hz

Q1.06.2

MCQ

2006

1M

Ans: C

● ○ ○ ○

Given: 2 V DC voltage and 4 V peak-to-peak square wave signal are superimposed i.e.

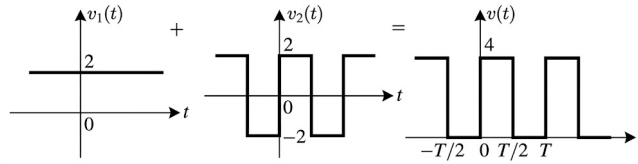


Fig. Solution Q1.06.2

RMS value of superimposed signal is given by,

$$v_{\text{rms}} = \sqrt{\frac{1}{T} \int_{-T/2}^{T/2} v^2(t) dt}$$

$$v_{\text{rms}} = \sqrt{\frac{1}{T} \left[\int_{-T/2}^0 0 dt + \int_0^{T/2} 4^2 dt \right]}$$

$$v_{\text{rms}} = \sqrt{\frac{1}{T} \left[16 \times \frac{T}{2} \right]} = \sqrt{8} \text{ V}$$

Hence, the correct option is (C).

(C) $\sqrt{8}$ V

Q1.05.1

MCQ

2005

1M

Ans: A

● ○ ○ ○

Given:

$$x(n) = 3 \sin(1.3\pi n + 0.5\pi) + 5 \sin(1.2\pi n)$$

$$\omega_1 = 1.3\pi \text{ rad/sec}, \quad \omega_2 = 1.2\pi \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2)$$

$$\omega_0 = \text{HCF}(1.3\pi, 1.2\pi)$$

$$\omega_0 = \text{HCF}\left(\frac{13\pi}{10}, \frac{12\pi}{10}\right)$$

$$\omega_0 = \frac{\text{HCF}[12\pi, 13\pi]}{\text{LCM}[10, 10]}$$

$$\omega_0 = \frac{\pi}{10} \text{ rad/sec}$$

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi/10} = 20$$

Hence, the correct option is (A).

(A) 20

Key Points

- Fundamental angular frequency:

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3, \dots)$$

- Fundamental period:

$$T = \frac{2\pi}{\omega_0}$$

Q1.01.1

MCQ

2001

2M

Ans: D

☒ ☐ ☐

Power of a discrete-time signal is

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$

Option (A): $x(n) = u(n)$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N 1 = \lim_{N \rightarrow \infty} \frac{N+1}{2N+1} = \frac{1}{2}$$

Hence, unit step sequence is a power signal.

Option (B): $x(n) = e^{j\omega_0 n}$

$$\begin{aligned} P &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |e^{j\omega_0 n}|^2 \\ &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N 1 = 1 \end{aligned}$$

Hence, $e^{j\omega_0 n}$ is a power signal.

Option (C): A periodic sequence.

Every periodic sequence has finite average power. Hence, it is a power signal.

Option (D): $x(n) = nu(n)$

$$\begin{aligned} P &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=0}^N n^2 \\ &= \lim_{N \rightarrow \infty} \frac{N(N+1)(2N+1)}{6(2N+1)} \\ &= \lim_{N \rightarrow \infty} \frac{N(N+1)}{6} = \infty \end{aligned}$$

Thus, the unit ramp sequence is *not* a power signal.

(D) Unit ramp sequence

SOLUTIONS



Basics of Systems

Topics

1. Continuous-Time Convolution
2. Discrete-Time Convolution
3. Linear and Non-Linear Systems
4. Time Variant and Time-Invariant Systems
5. LTI Systems - Definition and Properties
6. Causal and Non-Causal Systems
7. Static and Dynamic Systems
8. Stable and Unstable Systems
9. Invertible and Non-Invertible Systems
10. Systems Involving Differential Equations

Q2.26.1**MCQ****2026****2M****Ans: A****(M)** $y(t) = x^2(t)$ **Linearity:**Let $x(t) = ax_1(t) + bx_2(t)$.

$$y(t) = [ax_1(t) + bx_2(t)]^2$$

$$= a^2x_1^2(t) + b^2x_2^2(t) + 2abx_1(t)x_2(t)$$

Since

$$[ax_1(t) + bx_2(t)]^2 \neq ax_1^2(t) + bx_2^2(t),$$

the system is non-linear.

Time Invariance:For $x_1(t) = x(t - t_0)$,

$$y_1(t) = [x(t - t_0)]^2$$

and

$$y(t - t_0) = [x(t - t_0)]^2.$$

Hence $y_1(t) = y(t - t_0)$, so the system is time-invariant.

$$(M) \rightarrow (IV)$$

(N) $y(t) = x(-t)$ **Linearity:**

$$y(t) = ax_1(-t) + bx_2(-t) = ay_1(t) + by_2(t)$$

Hence, the system is linear.

Time Invariance:For $x_1(t) = x(t - t_0)$,

$$y_1(t) = x(-t - t_0)$$

while

$$y(t - t_0) = x(-t + t_0).$$

Since

$$x(-t - t_0) \neq x(-t + t_0),$$

the system is time-varying.

$$(N) \rightarrow (III)$$

(O) $y(t) = |x(3 - t)|$ **Linearity:**

$$y(t) = |ax_1(3 - t) + bx_2(3 - t)|$$

but

$$|ax_1(3-t) + bx_2(3-t)| \neq a|x_1(3-t)| + b|x_2(3-t)|.$$

Hence, the system is non-linear.

Time Invariance:

For $x_1(t) = x(t - t_0)$,

$$y_1(t) = |x(3 - t - t_0)|$$

while

$$y(t - t_0) = |x(3 - t + t_0)|.$$

Since

$$|x(3 - t - t_0)| \neq |x(3 - t + t_0)|,$$

the system is time-varying.

$$(O) \rightarrow (II)$$

(P) $y(t) = x(t - 3)$

Linearity:

$$y(t) = ax_1(t - 3) + bx_2(t - 3) = ay_1(t) + by_2(t)$$

Hence, the system is linear.

Time Invariance:

For $x_1(t) = x(t - t_0)$,

$$y_1(t) = x(t - t_0 - 3)$$

and

$$y(t - t_0) = x((t - t_0) - 3) = x(t - t_0 - 3).$$

Hence $y_1(t) = y(t - t_0)$, so the system is time-invariant.

$$(P) \rightarrow (I)$$

Therefore,

$$M-IV, \quad N-III, \quad O-II, \quad P-I$$

$$(A) \text{ M-IV, N-III, O-II, P-I}$$

Q2.25.1

MSQ

2025

2M

Ans: A,C



For an LTI system, an eigenfunction satisfies

$$T\{x(t)\} = \lambda x(t)$$

where λ is a scalar.

The eigenfunctions of continuous-time LTI systems are complex exponentials of the form

$$x(t) = e^{st}.$$

Option (A):

$$x(t) = e^{j\frac{2\pi}{3}t}$$

This is a complex exponential. Hence, it is an eigenfunction.

Option (B):

$$x(t) = \cos\left(\frac{2\pi}{3}t\right)$$

A cosine is a sum of two complex exponentials and is not an eigenfunction by itself.

Option (C):

$$x(t) = 2^t = e^{(\ln 2)t}$$

This is an exponential function of the form e^{st} . Hence, it is an eigenfunction.

Option (D):

$$x(t) = \sin\left(\frac{2\pi}{3}t\right)$$

A sine is also a combination of complex exponentials and is not an eigenfunction by itself.

(A), (C)

Q2.25.2

MCQ

2025

2M

Ans: C



Given,

$$y[n] = \frac{1}{\alpha}y[n-1] + x[n], \quad \alpha > 1$$

and

$$x[n] = \delta[n-p], \quad p > 10.$$

Substituting $x[n]$,

$$y[n] = \frac{1}{\alpha}y[n-1] + \delta[n-p]$$

For $n = p$,

$$y[p] = \frac{1}{\alpha}y[p-1] + \delta[0]$$

$$y[p] = \frac{1}{\alpha}y[p-1] + 1$$

For $n = p + 1$,

$$y[p+1] = \frac{1}{\alpha}y[p] + \delta[1]$$

$$y[p+1] = \frac{1}{\alpha}y[p]$$

Substituting $y[p]$,

$$y[p+1] = \frac{1}{\alpha} \left(\frac{1}{\alpha}y[p-1] + 1 \right)$$

$$y[p+1] = \frac{1}{\alpha^2}y[p-1] + \frac{1}{\alpha}$$

Since the system is initially at rest and $p > 10$,

$$y[p - 1] = 0$$

Therefore,

$$y[p + 1] = \frac{1}{\alpha}$$

$$\boxed{(C) \frac{1}{\alpha}}$$

Q2.22.1

MCQ

2022

1M

Ans: D



Given,

$$y(t) = x(t) \sin(2\pi t)$$

Linearity:

For inputs $x_1(t)$ and $x_2(t)$,

$$y_1(t) = x_1(t) \sin(2\pi t), \quad y_2(t) = x_2(t) \sin(2\pi t)$$

Then,

$$ay_1(t) + by_2(t) = ax_1(t) \sin(2\pi t) + bx_2(t) \sin(2\pi t)$$

$$= [ax_1(t) + bx_2(t)] \sin(2\pi t)$$

For input

$$x_3(t) = ax_1(t) + bx_2(t),$$

the output is

$$y_3(t) = [ax_1(t) + bx_2(t)] \sin(2\pi t).$$

Hence,

$$y_3(t) = ay_1(t) + by_2(t)$$

and the system is linear.

Time Invariance:

For delayed input $x(t - t_0)$,

$$y_1(t) = x(t - t_0) \sin(2\pi t)$$

The delayed version of the original output is

$$y(t - t_0) = x(t - t_0) \sin(2\pi(t - t_0))$$

Since

$$x(t - t_0) \sin(2\pi t) \neq x(t - t_0) \sin(2\pi(t - t_0)),$$

we have

$$y_1(t) \neq y(t - t_0).$$

Therefore, the system is time-varying.

(D) Linear and time-variant

Q2.19.1

MCQ

2019

1M

Ans: A

☒ ☐ ☐

Given,

$$y[n] = \alpha y[n-1] + x[n]$$

Taking the z -transform on both sides,

$$Y(z) = \alpha z^{-1} Y(z) + X(z)$$

$$Y(z) (1 - \alpha z^{-1}) = X(z)$$

Hence, the transfer function is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - \alpha z^{-1}}$$

The pole is located at

$$z = \alpha.$$

For a causal discrete-time system to be BIBO stable, all poles must lie inside the unit circle. Therefore,

$$|\alpha| < 1.$$

(A) $|\alpha| < 1$

Q2.14.1

MCQ

2014

1M

Ans: A

☒ ☐ ☐

Given:

$$h(n) = \begin{cases} \frac{\omega_c}{\pi}, & n = 0 \\ \frac{\sin(\omega_c n)}{\pi n}, & n \neq 0 \end{cases}$$

Condition for causal system:

Impulse response of the system should satisfy

$$h(n) = 0, \quad n < 0$$

Since,

$$h(n) = \frac{\sin(\omega_c n)}{\pi n}, \quad n \neq 0$$

the impulse response is non-zero for $n < 0$.

Therefore, the given system is a non-causal system.

$$h(n) = \frac{\sin(\omega_c n)}{\pi n}$$

Discrete-time Fourier transform of $h(n)$ is shown below.

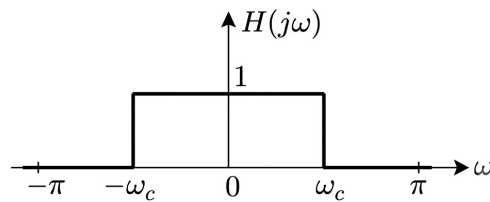


Fig. Solution Q2.14.1

Above figure represents an ideal low-pass filter.

(A) Non-causal, low-pass filter

Q2.13.1

MCQ

2013

1M

Ans: C



For z -plane, condition for stability for a causal system is that all the poles must lie inside

$$|z| = 1.$$

For s -plane, condition for stability for a causal system is that all the poles must lie in the negative half of s -plane.

Statement 1: All the poles of the system must lie on the left side of the $j\omega$ axis.

From above discussion, we can clearly show that statement 1 is true.

Statement 2: Zeros of the system can lie anywhere in the s -plane.

Since stability of the system depends on the poles, hence zeros can lie anywhere in s -plane for a stable system.

Statement 3: All the poles must lie within $|s| = 1$.

There is no such condition for stability in s -plane.

Therefore, statement 3 is wrong.

Statement 4: All the roots of the characteristic equation must be located on the left side of the $j\omega$ axis.

From above discussion, statement 4 is true.

Hence, the correct option is (C).

(C) All the poles must lie within $|s| = 1$.

Q2.13.2

MCQ

2013

2M

Ans: B



Given: Impulse response,

$$h(t) = \delta(t - 1) + \delta(t - 3)$$

$$\frac{d}{dt}(\text{Step response}) = \text{Impulse response}$$

$$\frac{d}{dt}[s(t)] = h(t)$$

Thus,

$$s(t) = \int_{-\infty}^t h(t) dt$$

$$s(t) = \int_{-\infty}^t [\delta(t - 1) + \delta(t - 3)] dt$$

$$s(t) = u(t - 1) + u(t - 3)$$

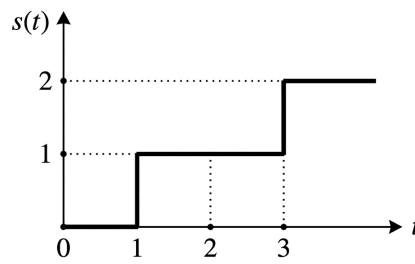


Fig. Solution Q2.13.2

$$s(t) \Big|_{t=2} = 1$$

Hence, the correct option is (B).

(B) 1

Q2.12.1

MCQ

2012

2M

Ans: D

☒ ☐ ☐

Given,

$$y(t) = \int_{-\infty}^t x(\tau) \cos(3\tau) d\tau$$

(i) **Time invariance:**

Let the input be delayed by t_0 . Then,

$$y(t, t_0) = \int_{-\infty}^t x(\tau - t_0) \cos(3\tau) d\tau$$

Let

$$\tau - t_0 = \tau_1$$

Then,

$$y(t, t_0) = \int_{-\infty - t_0}^{t - t_0} x(\tau_1) \cos[3(t_0 + \tau_1)] d\tau_1 \quad (i)$$

Replacing t by $t - t_0$,

$$y(t - t_0) = \int_{-\infty}^{t - t_0} x(\tau) \cos(3\tau) d\tau \quad (ii)$$

From (i) and (ii),

$$y(t, t_0) \neq y(t - t_0)$$

Hence, the system is time-varying.

(ii) **Stability:**

Let

$$x(t) = \cos(3t)$$

which is a bounded input.

Then,

$$y(t) = \int_{-\infty}^t \cos^2(3\tau) d\tau$$

Using

$$\cos^2(3\tau) = \frac{1 + \cos(6\tau)}{2},$$

$$y(t) = \int_{-\infty}^t \frac{1 + \cos(6\tau)}{2} d\tau$$

$$y(t) = \frac{1}{2} \left[\tau + \frac{\sin(6\tau)}{6} \right]_{-\infty}^t = \infty$$

Thus, for a bounded input, the output is unbounded.

Hence, the system is not stable.

(D) not time-invariant and not stable

Q2.11.1

MCQ

2011

1M

Ans: D



Given,

$$y(t) = \frac{d}{dt} \{e^{-t} x(t)\}$$

(i) Linearity:

For input $x_1(t)$,

$$y_1(t) = \frac{d}{dt} \{e^{-t} x_1(t)\}$$

Multiplying by scalar a ,

$$a y_1(t) = a \frac{d}{dt} \{e^{-t} x_1(t)\} \quad (i)$$

For input $x_2(t)$,

$$y_2(t) = \frac{d}{dt} \{e^{-t} x_2(t)\}$$

Multiplying by scalar b ,

$$b y_2(t) = b \frac{d}{dt} \{e^{-t} x_2(t)\} \quad (ii)$$

Adding (i) and (ii),

$$a y_1(t) + b y_2(t) = a \frac{d}{dt} \{e^{-t} x_1(t)\} + b \frac{d}{dt} \{e^{-t} x_2(t)\} \quad (iii)$$

Now let the input be

$$a x_1(t) + b x_2(t)$$

with output $y_3(t)$.

$$y_3(t) = \frac{d}{dt} \{e^{-t} (a x_1(t) + b x_2(t))\}$$

$$y_3(t) = a \frac{d}{dt} \{e^{-t} x_1(t)\} + b \frac{d}{dt} \{e^{-t} x_2(t)\} \quad (iv)$$

From (iii) and (iv),

$$y_3(t) = a y_1(t) + b y_2(t)$$

Therefore, the system satisfies superposition and is **linear**.

(ii) Time Variance:

For delayed input $x(t - t_0)$,

$$y_1(t) = \frac{d}{dt} \{e^{-t} x(t - t_0)\} \quad (v)$$

The delayed version of the output is

$$y(t - t_0) = \frac{d}{dt} \{e^{-(t-t_0)} x(t - t_0)\} \quad (vi)$$

From (v) and (vi),

$$y_1(t) \neq y(t - t_0)$$

Hence, the system is **time-varying**.

(iii) Stability:

For bounded input

$$x(t) = u(t),$$

$$y(t) = \frac{d}{dt} \{e^{-t} u(t)\}$$

$$y(t) = e^{-t} \delta(t) + u(t)(-e^{-t})$$

For bounded input, the output is unbounded due to the impulse term.

Therefore, the system is **unstable**.

(iv) Memory:

A system is memoryless if the output depends only on the present input.

Differentiation depends on neighboring values of the input signal.

Hence, the system possesses **memory** and is a dynamic system.

(D) The system has memory

Q2.10.1

MCQ

2010

2M

Ans: C

● ○ ○

Given,

$$y(t) = \int_{-\infty}^{5t} x(\tau) d\tau$$

(i) Time Variance:

Let the input be delayed by t_0 . Then the corresponding output is

$$y_1(t) = \int_{-\infty}^{5t} x(\tau - t_0) d\tau \quad (i)$$

Now delay the output $y(t)$ by t_0 ,

$$y(t - t_0) = \int_{-\infty}^{5(t-t_0)} x(\tau) d\tau \quad (ii)$$

From (i) and (ii),

$$y_1(t) \neq y(t - t_0)$$

Therefore, the system is time-varying.

(ii) Causality:

If the output of a system depends only on present and past inputs, then the system is causal. If it depends on future inputs also, then it is non-causal.

Given,

$$y(t) = \int_{-\infty}^{5t} x(\tau) d\tau$$

For $t = -1$,

$$y(-1) = \int_{-\infty}^{-5} x(\tau) d\tau$$

For $t = 1$,

$$y(1) = \int_{-\infty}^5 x(\tau) d\tau$$

Here, it is clear that the output depends upon both past and future values of the input. Therefore, the system is non-causal.

(C) time variant and non-causal

Q2.09.1

MCQ

2009

2M

Ans: D



Given,

$$y(t) = \sum_{k=-\infty}^{\infty} x(kT) \delta(t - kT)$$

(i) Linearity:

For input $x_1(t)$, output will be $y_1(t)$.

$$y_1(t) = \sum_{k=-\infty}^{\infty} x_1(kT) \delta(t - kT)$$

Multiplying both sides by scalar quantity a ,

$$a y_1(t) = a \sum_{k=-\infty}^{\infty} x_1(kT) \delta(t - kT) \quad (i)$$

For input $x_2(t)$, output will be $y_2(t)$.

$$y_2(t) = \sum_{k=-\infty}^{\infty} x_2(kT) \delta(t - kT)$$

Multiplying both sides by scalar quantity b ,

$$b y_2(t) = b \sum_{k=-\infty}^{\infty} x_2(kT) \delta(t - kT) \quad (ii)$$

Adding equations (i) and (ii),

$$a y_1(t) + b y_2(t) = \sum_{k=-\infty}^{\infty} [a x_1(kT) + b x_2(kT)] \delta(t - kT) \quad (iii)$$

Now, let the system have input

$$a x_1(t) + b x_2(t)$$

and corresponding output be $y_3(t)$.

$$y_3(t) = \sum_{k=-\infty}^{\infty} [a x_1(kT) + b x_2(kT)] \delta(t - kT) \quad (\text{iv})$$

From equations (iii) and (iv),

$$y_3(t) = a y_1(t) + b y_2(t)$$

Therefore, the system follows the principle of superposition. Hence, the system is linear.

(ii) Time Variance:

In a time invariant system, any delay provided in input must be reflected in output.

Let the input be delayed by t_0 , then corresponding output is $y_1(t)$.

$$y_1(t) = \sum_{k=-\infty}^{\infty} x(kT - t_0) \delta(t - kT) \quad (\text{v})$$

Now, output $y(t)$ is delayed by t_0 ,

$$y(t - t_0) = \sum_{k=-\infty}^{\infty} x(kT) \delta(t - t_0 - kT) \quad (\text{vi})$$

From equations (v) and (vi),

$$y_1(t) \neq y(t - t_0)$$

Thus, the system is time varying.

(D) linear and time-varying

Q2.15.1

MCQ

2015

1M

Ans: D



Time invariance:

In a time invariant system, any delay provided in input must be reflected in output.

(i) From option (A): $y(n) = n x(n)$

Let input be delayed by n_0 . Then corresponding output is

$$y_1(n) = n x(n - n_0) \quad (\text{i})$$

Now, output $y(n)$ is delayed by n_0 ,

$$y(n - n_0) = (n - n_0)x(n - n_0) \quad (\text{ii})$$

From equations (i) and (ii),

$$y_1(n) \neq y(n - n_0)$$

Thus, the system is time varying.

(ii) From option (B): $y(n) = x(3n)$

Let input be delayed by n_0 . Then corresponding output is

$$y_1(n) = x(3n - n_0) \quad (\text{iii})$$

Now, output $y(n)$ is delayed by n_0 ,

$$y(n - n_0) = x[3(n - n_0)] \quad (\text{iv})$$

From equations (iii) and (iv),

$$y_1(n) \neq y(n - n_0)$$

Thus, the system is time varying.

(iii) From option (C): $y(n) = x(-n)$

Let input be delayed by n_0 . Then corresponding output is

$$y_1(n) = x(-n - n_0) \quad (v)$$

Now, output $y(n)$ is delayed by n_0 ,

$$y(n - n_0) = x[-(n - n_0)] \quad (vi)$$

From equations (v) and (vi),

$$y_1(n) \neq y(n - n_0)$$

Thus, the system is time varying.

(iv) From option (D): $y(n) = x(n - 3)$

Let input be delayed by n_0 . Then corresponding output is

$$y_1(n) = x(n - n_0 - 3) \quad (vii)$$

Now, output $y(n)$ is delayed by n_0 ,

$$y(n - n_0) = x[(n - 3) - n_0] \quad (viii)$$

From equations (vii) and (viii),

$$y_1(n) = y(n - n_0)$$

Thus, the system is time invariant.

(D) $y[n] = x[n - 3]$

Q2.01.1

MCQ

2001

1M

Ans: D



Condition for causality of a LTI system:

$$h(n) = 0; \quad n < 0$$

(i) From option (A): $h(n) = a^n u(n - 2)$

This signal will exist only for $n \geq 2$ i.e., for $n < 0$, $h(n) = 0$.

Therefore, it is causal.

(ii) From option (B): $h(n) = a^{n-2} u(n)$

This signal will exist only for $n \geq 0$ i.e., for $n < 0$, $h(n) = 0$.

Therefore, it is causal.

(iii) From option (C): $h(n) = a^{n+2} u(n)$

This signal will exist only for $n \geq 0$ i.e., for $n < 0$, $h(n) = 0$.

Therefore, it is causal.

(iv) From option (D): $h(n) = a^n u(n + 2)$

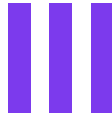
This signal exists only for $n \geq -2$ i.e., for $n < 0$, $h(n) \neq 0$.

Therefore, it is non-causal.

Hence, the correct option is (D).

(D) $a^n u(n + 2)$

SOLUTIONS



Z Transform

Topics

1. Introduction to Z-Transform
2. Z-Transform and ROC
3. Inverse Z-Transform
4. Important Properties of Z-Transform
5. Unilateral Z-Transform

Q6.24.1 NAT 2024 2M Ans: 0.75 ☐ ☐ ☐

Given $y[n] - 0.5y[n-2] = ax[n-4]$. Taking the Z-transform,

$$Y(z) - 0.5z^{-2}Y(z) = aZ^{-4}X(z) \Rightarrow H(Z) = \frac{Y(Z)}{X(Z)} = \frac{aZ^{-4}}{1 - 0.5Z^{-2}}.$$

Substituting $z = e^{j\omega}$ for the frequency response,

$$H(e^{j\omega}) = \frac{ae^{-j4\omega}}{1 - 0.5e^{-j2\omega}}.$$

At $\omega = \frac{\pi}{2}$, given $|H(e^{j\pi/2})| = 0.5$:

$$0.5 = \left| \frac{ae^{-j2\pi}}{1 - 0.5e^{-j\pi}} \right| = \frac{a}{1 - 0.5(-1)} = \frac{a}{1.5}.$$

$$a = 1.5 \times 0.5 = 0.75.$$

Hence, the value of a is **0.75**.

Q6.24.2 MCQ 2024 2M Ans: C ☒ ☒ ☐

Given

$$x(n) = \begin{cases} (0.2)^n, & 0 \leq n \leq 7 \\ 0, & \text{otherwise} \end{cases}$$

$$X(z) = \sum_{n=0}^7 (0.2)^n z^{-n} = 1 + (0.2)z^{-1} + (0.2)^2 z^{-2} + \dots + (0.2)^7 z^{-7}.$$

This is a finite sum of negative powers of z . It converges everywhere *except* $z = 0$, where the negative-power terms are undefined; there is no restriction at $z = \infty$ since the expression reduces to the finite constant term 1 there.

Hence, the ROC is all values of z except $z = 0$, and the correct option is **(C)**.

Q6.23.1 MCQ 2023 2M Ans: B ☒ ☒ ☐

Given $F(z) = \frac{1}{1-z}$, expanded around $z = 2$. Let $t = z - 2$, so $z = 2 + t$.

$$F(t) = \frac{1}{1 - (2+t)} = \frac{1}{-1-t} = \frac{-1}{1+t} = -\left(1 - t + t^2 - t^3 + \dots\right).$$

$$F(z) = -1 + (z-2) - (z-2)^2 + (z-2)^3 - \dots = \sum_{k=0}^{\infty} (-1)^{k+1} (z-2)^k.$$

Comparing with $F(z) = \sum_{k=0}^{\infty} a_k (z-2)^k$, we get $a_k = (-1)^{k+1}$.

Hence, the correct option is **(B)**.

Q6.18.1 MCQ 2018 2M Ans: D ☒ ☒ ☐

Given $Y(z) = 2 + 3z^{-1} + z^{-2}$ is the Z-transform of $y[n] = x[n] * h[n]$, and $p[n] = x[n] * h[n-2]$.

By the time-shifting property, convolving with a signal delayed by 2 samples multiplies the transform by z^{-2} :

$$P(z) = z^{-2}Y(z) = z^{-2}(2 + 3z^{-1} + z^{-2}) = 2z^{-2} + 3z^{-3} + z^{-4}.$$

Hence, the correct option is **(D)**.

Q6.17.1 MCQ 2017 1M Ans: C ☒ ☐ ☐

Given $x_1(n) = \{1, 1\}$ and $x_2(n) = \{1, 2\}$.

$$x_1(n) \xrightarrow{\text{Z.T.}} 1 + z^{-1}, \quad x_2(n) \xrightarrow{\text{Z.T.}} 1 + 2z^{-1}.$$

By the convolution property,

$$x_1(n) \otimes x_2(n) = X_1(z)X_2(z) = (1 + z^{-1})(1 + 2z^{-1}) = 1 + 3z^{-1} + 2z^{-2}.$$

Hence, the correct option is (C).

Q6.15.1 MCQ 2015 1M Ans: A ☒ ☐ ☐

Given $x(n) = \alpha^{|n|}$, $0 < |\alpha| < 1$. This can be split as

$$x(n) = \underbrace{\alpha^n u(n)}_{x_1(n)} + \underbrace{\alpha^{-n} u(-n-1)}_{x_2(n)}.$$

Using the standard pairs,

$$\begin{aligned} \alpha^n u(n) &\xleftrightarrow{\text{ZT}} \frac{1}{1 - \alpha z^{-1}}, \quad \text{ROC : } |z| > |\alpha|, \\ -\left(\frac{1}{\alpha}\right)^n u(-n-1) &\xleftrightarrow{\text{ZT}} \frac{1}{1 - \frac{1}{\alpha} z^{-1}}, \quad \text{ROC : } |z| < \frac{1}{|\alpha|}. \end{aligned}$$

The common ROC exists (since $|\alpha| < 1$) and equals

$$|\alpha| < |z| < \frac{1}{|\alpha|}.$$

Hence, the correct option is (A).

Q6.14.1 MCQ 2014 2M Ans: C ☒ ☒ ☐

Given $\frac{b_0}{1 - z^{-1} + a_2 z^{-2}}$, a_2 real. Multiplying through by z^2 , the poles satisfy

$$z^2 - z + a_2 = 0 \Rightarrow z = \frac{1 \pm \sqrt{1 - 4a_2}}{2}.$$

Checking each option for $|z| < 1$ (BIBO stability):

- $a_2 = 0.5$: $z = \frac{1 \pm j}{2}$, $|z| = \frac{1}{\sqrt{2}} \approx 0.707 < 1 \Rightarrow$ **stable**.
- $a_2 = 1.5$: $|z| = \frac{\sqrt{6}}{2} \approx 1.22 > 1 \Rightarrow$ **unstable**.
- $a_2 = -0.75$: $z = 1.5, -0.5$; $|1.5| > 1 \Rightarrow$ **unstable**.
- $a_2 = -1.5$: $z \approx 1.82, -0.82$; $|1.82| > 1 \Rightarrow$ **unstable**.

Hence, the correct option is (C).

Q6.14.2**MCQ****2014****2M****Ans: C**

Given $H(z) = \frac{1 - \frac{1}{3}z^{-1}}{1 - \frac{1}{4}z^{-1}}$. $H(z)$ has a pole at $z = \frac{1}{4}$ and a zero at $z = \frac{1}{3}$.

For $H(z)$ itself to be causal and stable, its ROC must lie outside its outermost pole:

$$|z| > \frac{1}{4}.$$

Hence, the correct option is **(C)**.

Q6.13.1**MCQ****2013****1M****Ans: D**

Given $\frac{1 - 2z^{-1}}{1 - 0.5z^{-1}}$.

Zero: $1 - 2z^{-1} = 0 \Rightarrow z = 2$, which lies *outside* the unit circle $\Rightarrow$ the system is **non-minimum phase**.

Pole: $1 - 0.5z^{-1} = 0 \Rightarrow z = 0.5$, which lies *inside* the unit circle $\Rightarrow$ the system is **stable**.

Hence, the correct option is **(D)**.

Q6.12.1**MCQ****2012****2M****Ans: C**

Given $x[n] = \left(\frac{1}{3}\right)^{|n|} - \left(\frac{1}{2}\right)^n u[n]$.

The two-sided term splits as $\left(\frac{1}{3}\right)^{|n|} = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{3}\right)^{-n} u[-n-1]$, whose ROC is $\frac{1}{3} < |z| < 3$.

The term $-\left(\frac{1}{2}\right)^n u[n]$ is right-sided with ROC $|z| > \frac{1}{2}$.

The overall ROC is the intersection of the two:

$$\max\left(\frac{1}{3}, \frac{1}{2}\right) < |z| < 3 \Rightarrow \frac{1}{2} < |z| < 3.$$

Hence, the correct option is **(C)**.

Q6.11.1**MCQ****2011****2M****Ans: B**

Given $y(n) - \frac{1}{3}y(n-1) = x(n)$, $x(n) = \left(\frac{1}{2}\right)^n u(n)$. Taking the Z-transform under initial-rest conditions,

$$Y(z) - \frac{1}{3}z^{-1}Y(z) = \frac{z}{z - \frac{1}{2}} \Rightarrow Y(z) = \frac{z}{\left(z - \frac{1}{2}\right)\left(1 - \frac{1}{3}z^{-1}\right)} = \frac{z^2}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{3}\right)}.$$

Expanding $Y(z)/z$ in partial fractions,

$$\frac{Y(z)}{z} = \frac{3}{z - \frac{1}{2}} - \frac{2}{z - \frac{1}{3}} \Rightarrow Y(z) = \frac{3z}{z - \frac{1}{2}} - \frac{2z}{z - \frac{1}{3}}.$$

Taking the inverse Z-transform,

$$y(n) = -2\left(\frac{1}{3}\right)^n + 3\left(\frac{1}{2}\right)^n, \quad n \geq 0.$$

Hence, the correct option is **(B)**.

Q6.10.1 **MCQ** **2010** **2M** **Ans: B** ● ● ○

For $H(z)$ to be stable, its poles must lie inside the unit circle. For the inverse system $1/H(z)$ to be stable, its poles — i.e., the zeros of $H(z)$ — must also lie inside the unit circle.

Hence both the poles and zeros of $H(z)$ must lie inside the unit circle, and the correct option is **(B)**.

Q6.08.1 **MCQ** **2008** **1M** **Ans: A** ● ○ ○

Given $x(n) = 2^n u(n)$, a right-sided (causal) signal.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = \sum_{n=0}^{\infty} 2^n z^{-n} = \sum_{n=0}^{\infty} (2z^{-1})^n.$$

This converges only if $|2z^{-1}| < 1$, i.e. $|z| > 2$, giving

$$X(z) = \frac{1}{1 - 2z^{-1}} = \frac{z}{z - 2}, \quad \text{ROC : } |z| > 2.$$

Hence, the correct option is **(A)**.

Q6.08.2 **MCQ** **2008** **2M** **Ans: A** ● ● ○

Given $y(n) = x(n) - \frac{1}{3}y(n-1)$, $x(n) = \delta(n)$, with $y(n) = 0$ for $n < 0$ (causal). Taking the Z-transform,

$$Y(z) = 1 - \frac{1}{3}z^{-1}Y(z) \Rightarrow Y(z)\left(1 + \frac{1}{3z}\right) = 1 \Rightarrow Y(z) = \frac{3z}{3z + 1} = \frac{z}{z + \frac{1}{3}}.$$

Taking the inverse Z-transform,

$$y(n) = \left(-\frac{1}{3}\right)^n u(n).$$

Hence, the correct option is **(A)**.

Q6.05.1 **MCQ** **2005** **2M** **Ans: D** ● ● ○

From the given block diagram,

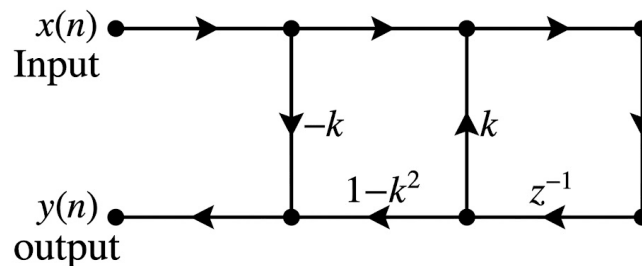


Fig. Solution Q6.05.1

$$W(z) = X(z) + z^{-1}k W(z) \Rightarrow W(z)(1 - kz^{-1}) = X(z) \Rightarrow W(z) = \frac{X(z)}{1 - kz^{-1}}. \quad (i)$$

Also,

$$Y(z) = -kX(z) + (1 - k^2)z^{-1}W(z).$$

Substituting (i),

$$\frac{Y(z)}{X(z)} = -k + \frac{(1 - k^2)z^{-1}}{1 - kz^{-1}} = \frac{-k + k^2z^{-1} + z^{-1} - k^2z^{-1}}{1 - kz^{-1}} = \frac{-k + z^{-1}}{1 - kz^{-1}}.$$

Hence, the correct option is **(D)**.

Q6.04.1

MCQ

2004

2M

Ans: C



Given:

$$X(z) = \frac{1/2}{1 - az^{-1}} + \frac{1/3}{1 - bz^{-1}} \quad \dots (i)$$

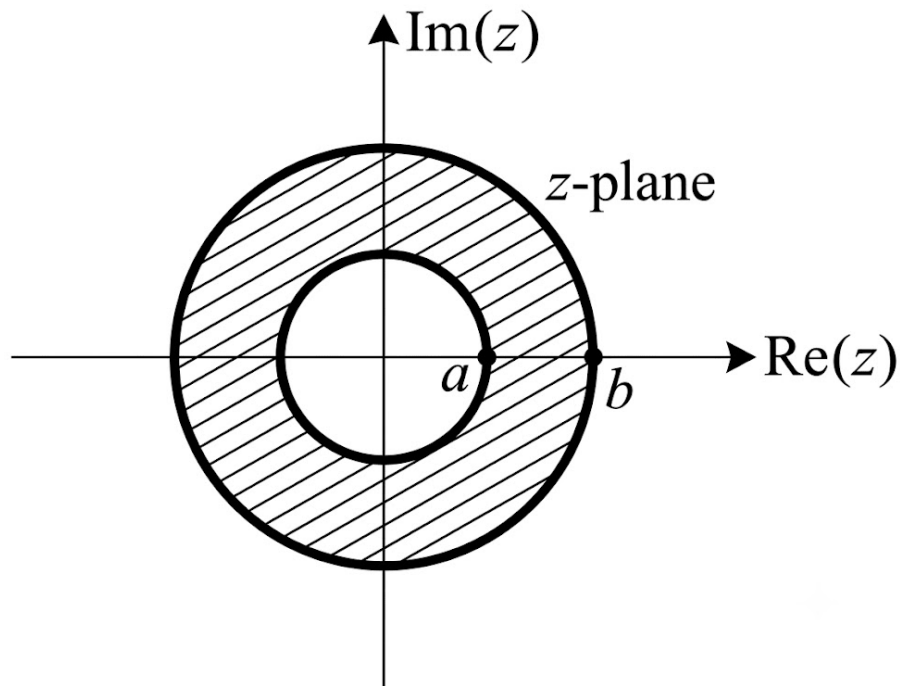
where $|a| < 1$ and $|b| < 1$ ROC : $|a| < |z| < |b|$ 

Fig. Solution Q6.04.1

According to given ROC,

$$X(z) = \underbrace{\frac{(1/2)z}{z-a}}_{\text{Right sided signal}} + \underbrace{\left(\frac{1}{3}\right)\frac{z}{z-b}}_{\text{Left sided signal}}$$

$$x(n) = \frac{1}{2}(a)^n u(n) - \frac{1}{3}(b)^n u(-n-1)$$

$$x(0) = \frac{1}{2}$$

$$\boxed{(C) \frac{1}{2}}$$

Key Points

- An ROC of the form $|a| < |z| < |b|$ represents a two-sided sequence consisting of a right-sided part for $|z| > |a|$ and a left-sided part for $|z| < |b|$.
- The inverse z-transform yields $x(0) = \frac{1}{2}$ since $u(0) = 1$ and $u(-1) = 0$.

Q6.04.2

MCQ

2004

1M

Ans: A



Given $H(z) = \frac{1}{1 - 0.7z^{-1} + 0.13z^{-2}}$, realized in direct form with feed-forward gain a_0 and feedback gains a_1, a_2 as shown in the figure.

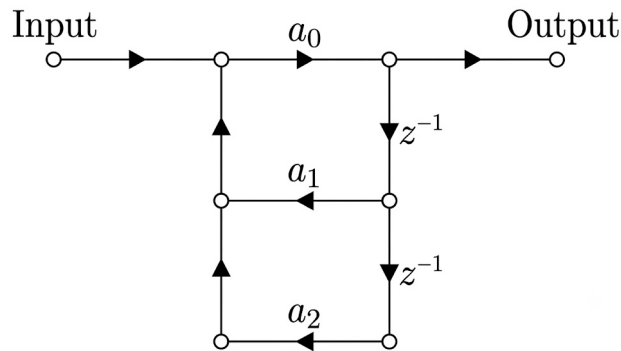


Fig. Solution Q6.04.2

Comparing the denominator coefficients of $H(z)$ with the direct-form structure,

$$a_0 = 1.0, \quad a_1 = 0.7, \quad a_2 = -0.13.$$

Hence, the correct option is (A).

Q6.04.3

MCQ

2004

2M

Ans: D



Given: Discrete time sequence $x(n)$ suffered distortion modeled by an LTI system with,

$$H(z) = (1 - az^{-1}); \quad |a| > 1$$

To compensate the distortion, the system transfer function of a stable system should be $\frac{1}{H(z)}$.

$$\frac{1}{H(z)} = \frac{1}{1 - az^{-1}}$$

Inverse transform is given by,

For right sided signal,

$$h_c(n) = a^n u(n); \quad \text{ROC} : |z| > |a| \quad \dots (i)$$

For left sided signal,

$$h_c(n) = -a^n u(-n - 1); \quad \text{ROC} : |z| < |a| \quad \dots (ii)$$

For a stable system, the ROC must include the unit circle ($|z| = 1$). Since $|a| > 1$, the ROC $|z| < |a|$ includes the unit circle, so equation (ii) represents the stable impulse response.

(D) $-a^n u(-n - 1)$

Key Points

- An LTI system is BIBO stable if and only if the Region of Convergence (ROC) of its system function $H(z)$ contains the unit circle ($|z| = 1$).
- When a pole lies outside the unit circle ($|a| > 1$), a stable system requires a left-sided impulse response with ROC $|z| < |a|$.

Q6.03.1

MCQ

2003

1M

Ans: A

● ○ ○

Given $X(z) = e^{1/z}$. Expanding as a power series in z^{-1} ,

$$X(z) = e^{1/z} = 1 + \frac{z^{-1}}{1!} + \frac{z^{-2}}{2!} + \frac{z^{-3}}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}. \quad (i)$$

By definition,

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}. \quad (ii)$$

Comparing (i) and (ii):

$$x(n) = \frac{1}{n!} \text{ for } n \geq 0 \quad \Rightarrow \quad x(n) = \frac{u(n)}{n!}.$$

Hence, the correct option is (A).

Q6.03.2

MCQ

2003

2M

Ans: A

● ● ○

Given input $x = [a, b, c, d]$ and output $y = [x, x, x, x, \dots \langle \text{repeated } N \text{ times} \rangle]$, i.e.

$$x(n) = a\delta(n) + b\delta(n-1) + c\delta(n-2) + d\delta(n-3),$$

$$y(n) = a\delta(n) + b\delta(n-1) + c\delta(n-2) + d\delta(n-3) + a\delta(n-4) + b\delta(n-5) + \cdots$$

The output repeats after every 4th sample, so the fundamental period is $N = 4$, and

$$y(n) = \sum_{i=0}^{N-1} x(n-4i) \quad \Rightarrow \quad Y(z) = \sum_{i=0}^{N-1} X(z) z^{-4i}.$$

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{i=0}^{N-1} z^{-4i} = 1 + z^{-4} + z^{-8} + \cdots + z^{-4(N-1)},$$

$$h(n) = \sum_{i=0}^{N-1} \delta(n-4i).$$

Per the official Answer Key, the correct option is (D).

Q6.01.1

MCQ

2001

1M

Ans: A

● ○ ○

Given $x(n) = \delta(n) \xrightarrow{h(n)} y(n)$. By the convolution property, $y(n) = x(n) \otimes h(n) \xrightarrow{\text{Z.T.}} Y(z) = X(z)H(z)$. With $x(n) = \delta(n) \leftrightarrow 1$ and $h(n) = a^n u(n) \leftrightarrow \frac{z}{z-a}$,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z}{z-a}.$$

For a unit-step input $x_1(n) = u(n) \leftrightarrow \frac{z}{z-1}$, ROC : $|z| > 1$.

$$Y_1(z) = \frac{z}{z-1} \cdot \frac{z}{z-a} = \frac{1}{1-a} \cdot \frac{z}{z-1} + \frac{a}{a-1} \cdot \frac{z}{z-a}.$$

Taking the inverse Z-transform,

$$y(n) = \frac{1}{1-a} u(n) + \frac{a}{a-1} a^n u(n) = \sum_{j=0}^n a^j.$$

Hence, the correct option is (A).